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1. Introduction & Module Architecture

The Learning & Development (L&D) module is 3Boxes HRMS's digital engine for employee upskilling, compliance training and professional growth. It replaces the familiar spreadsheet-plus-email training process with a single automated workflow: skill gaps are identified from performance reviews, courses are nominated and approved, enrolments are created with deadlines, delivery and assessments are tracked live, digital certificates are issued automatically on passing, and the entire cycle is measured through the Kirkpatrick four-level evaluation model ending in a financial ROI figure. Because every step writes an auditable record, the module also doubles as the company's compliance evidence trail for POSH, cybersecurity and safety mandates.

The module lives in the main sidebar under the **Learning & Development** section and is organised into eight pages: **L&D Hub**, **Course Catalog**, **Learning Paths**, **Assignments & Enrolment**, **Course Requests**, **Certifications**, **Compliance Center** and **Analytics & ROI**. Every employee in the organisation can see the section; the pages themselves adapt to the signed-in role. Administrators get management consoles and KPI cards, managers get team approvals and result recording, and employees get their personal learning plan, certificates and signature queue.

> **About the screenshots in this manual**
>
> All screenshots in this manual were captured on the 3Boxes HRMS demo tenant (nexus-hrms-mu.vercel.app) with sample Indian-rupee data pre-loaded for evaluation. Figures such as course costs, scores and compliance percentages are demonstration values, not production data.

1.1 What the module manages

The L&D module is built around eleven linked data records that together cover the full training lifecycle. Understanding them helps every other chapter of this manual make sense, because each screen is a window onto one or two of these records. The records, and the screens that manage them, are summarised below.

Record	Purpose	Managed from
Course + Course Modules	The catalog entry: content hub with videos, documents, quizzes, external links, cost, skills and passing score.	Course Catalog
Learning Path (+ ordered courses)	A structured sequence of courses: onboarding, compliance bundles, role transitions and promotion tracks.	Learning Paths
Course Assignment	A bulk instruction to train individuals, a department or the whole company by a due date.	Assignments & Enrolment
Course Enrolment	One employee × one course: progress, due date, pre/post-test scores, feedback and result.	Assignments & Enrolment / L&D Hub
Certificate	Auto-issued on passing; carries a unique number, validity window and renewal-alert state.	Certifications
Training Request	An employee's nomination for a course (catalog or external) awaiting supervisor/L&D approval.	Course Requests
Compliance Requirement	A recurring mandate (POSH, cybersecurity, safety) linked to a course with a frequency.	Compliance Center
Policy Sign-off + Acknowledgement	A bulk digital-signature campaign for policies and handbooks, with timestamped acknowledgements.	Compliance Center
Skill Gap	A TNI record: the skill, current vs required level, source review and the suggested course.	Hub (TNI card) / APIs

Table 1-1: Core L&D records and where they are managed

1.2 The eight-step automated workflow

The module implements one continuous digital workflow, and the L&D Hub shows it as a strip so every user can see where a given enrolment currently sits. The workflow begins with Training Needs Identification (TNI) and ends only when the employee's skill record and the company's analytics reflect the completed learning. Each step is either fully automatic or a single-click action for the responsible role, which is what makes the process scalable across departments.

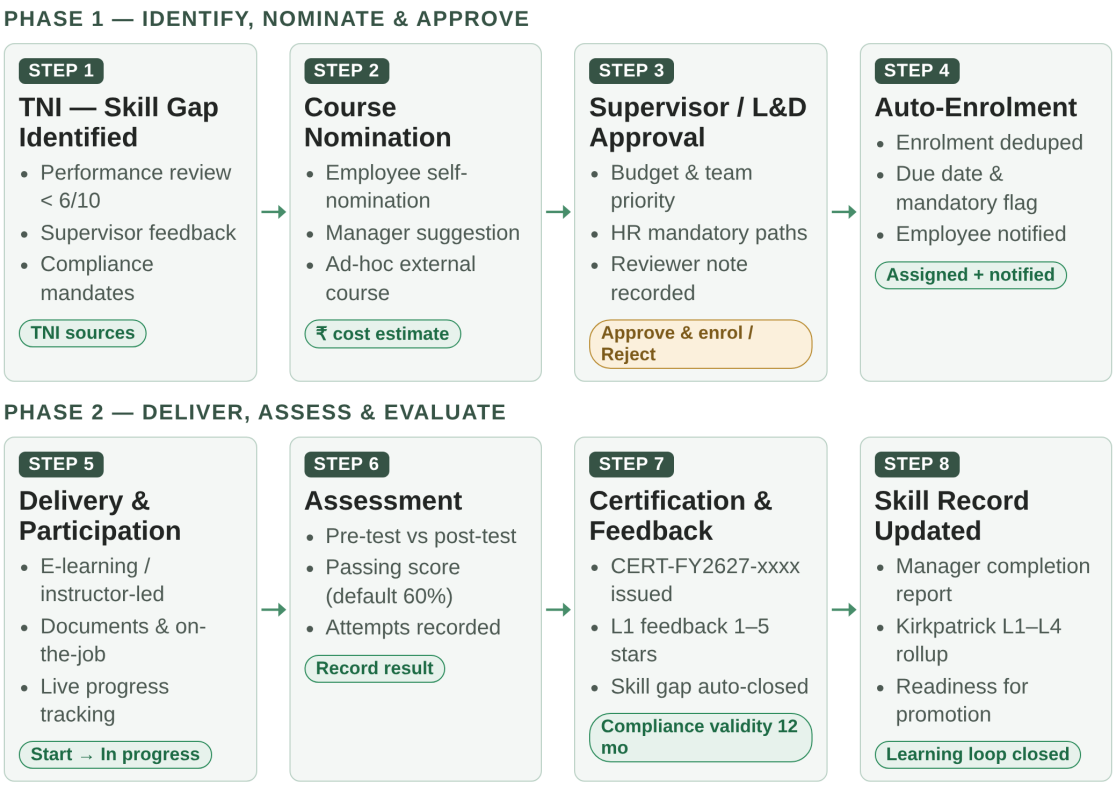


Exhibit 1-1: The end-to-end L&D; workflow — Phase 1 identifies, nominates and approves; Phase 2 delivers, assesses and evaluates

Steps 1–3 are covered in Chapter 6 (nominations and approvals) and Chapter 8 (TNI from performance reviews). Step 4 is the enrolment engine of Chapter 5. Steps 5–7 run on the enrolment lifecycle described in Chapter 7, and step 8 closes the loop by feeding the analytics of Chapter 10. The remainder of this chapter explains how to find your way around the module; later chapters are ordered the same way as the workflow itself.

1.3 Roles at a glance

Five roles use the module, and the platform gates every action to the appropriate one. Employees and managers see personal and team views; HR managers and administrators (tenant admin, admin, super admin) additionally manage the catalog, paths, compliance requirements and approvals. The complete permission matrix is in Chapter 12, but the practical split is worth memorising early: **employees request and learn**, **managers approve and record results**, and **L&D administrators build and govern**.

- **Employee** — sees My Learning, requests training, takes courses, signs policies, earns certificates.
- **Manager** — everything an employee sees, plus team approvals and the ability to record assessment results and suggest training.
- **HR Manager** — management actions on catalog, paths, assignments, approvals, compliance and sign-offs.
- **Admin / Tenant Admin / Super Admin** — full control including deletion of courses and paths, and the audit export.

2. L&D Hub — Your Starting Point

```
Navigation: Sidebar → Learning & Development → L&D Hub | Route: /learning
```

The L&D Hub is the role-aware landing page for the whole module. Its subtitle states the intent plainly: *"Upskilling, compliance tracking and professional growth through an automated digital workflow."* From here you can reach every other page, watch your personal learning queue, and — if you are an administrator — monitor the organisation's training posture through live KPI cards. The page is deliberately compact: it answers "what do I do next?" in one glance and defers detail to the dedicated pages.

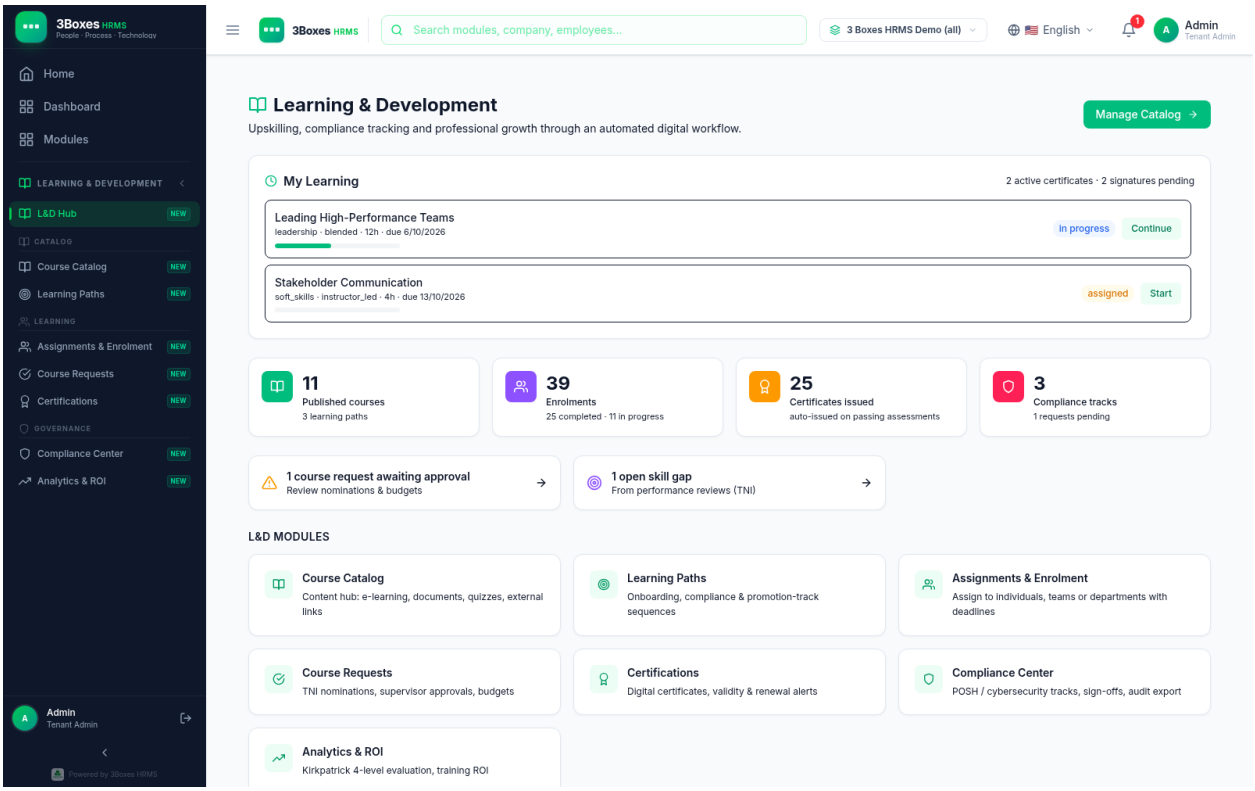


Exhibit 2-1: L&D; Hub (admin view) — KPI cards, alert cards and the L&D; Modules grid

2.1 Employee view — My Learning

For employees the most important block is **My Learning**. It lists the courses assigned to them with live progress bars, the due date, and a status badge (assigned, in progress, completed, failed or expired). The header line also summarises standing obligations, for example *"2 active certificates · 2 signatures pending"*, which counts active certificates and unsigned policy acknowledgements. When nothing is pending, the panel shows the friendly message *"No assigned courses pending — browse the catalog to request training."* and points employees at self-nomination. Each row carries a **Start** or **Continue** button that immediately moves the enrolment into in-progress state with a timestamp, so managers can see who has actually begun.

1. Sign in and open **L&D Hub** from the sidebar.
2. Review the **My Learning** rows — the highest-priority item is the one nearest its due date.

3. Click **Start** (first time) or **Continue** (returning) on a course row.
4. Complete the content, then wait for your manager or L&D admin to record the assessment result — a passed result automatically issues your certificate (Chapter 7).

2.2 Administrator view — KPI and alert cards

Administrators see four KPI cards across the top: **Published courses** (with the number of learning paths as the sub-label), **Enrolments** (split into completed and in progress), **Certificates issued** (auto-issued on passing assessments) and **Compliance tracks** (with pending training requests as the sub-label). Below them, up to three alert cards appear only when attention is genuinely needed: pending course requests awaiting approval, open skill gaps from performance reviews (TNI), and team requests a manager-direct supervisor still has to approve. Clicking any alert takes you straight to the queue that can clear it, which keeps the hub an action surface rather than a dashboard you must interpret.

The **L&D Modules** grid then links to the seven working pages with one-line descriptions, and admins also get a **Manage Catalog** shortcut in the header. The bottom of the page renders the **L&D Workflow** strip — the same eight steps shown in Exhibit 1-1 — so new users always have the process map one scroll away.

3. Course Catalog & Content Hub

```
Navigation: Sidebar → Learning & Development → Course Catalog | Route: /learning/catalog
```

The Course Catalog is the central repository for everything the company teaches. Each course card shows its delivery mode (e-learning, instructor-led, document, quiz, external link or blended), difficulty level, duration in hours, number of modules, cost in rupees, and live engagement counters (*"30 enrolled · 83% complete"*). Compliance courses carry an amber *Compliance · POSH*-style badge, and promotion-gate courses carry a blue *Promotion gate* badge — both flags drive behaviour elsewhere in the module, so they are set carefully at creation time.

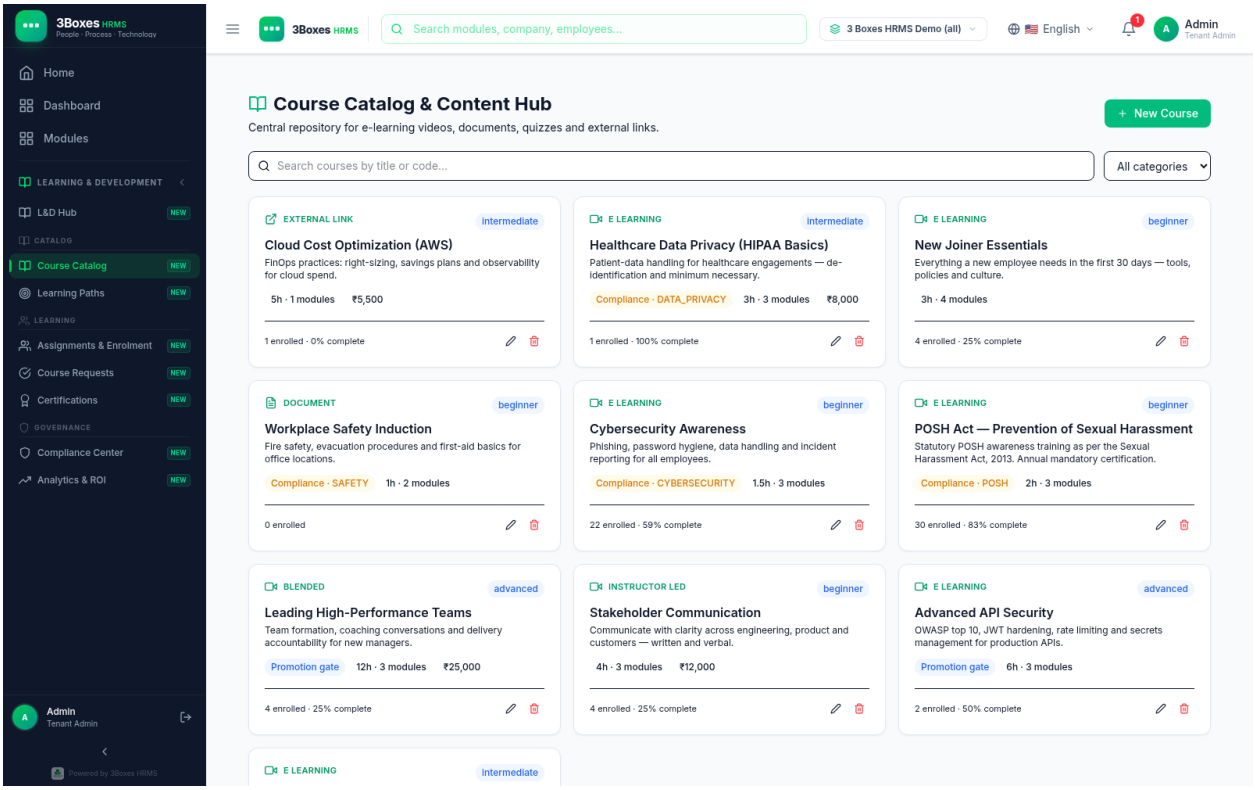


Exhibit 3-1: Course Catalog & Content Hub with sample INR-priced courses

3.1 Finding courses

The search box filters by title or code as you type (try *POSH-101*), and the **All categories** dropdown narrows the grid to one of six categories: technical, soft skills, compliance, leadership, onboarding or domain. Category filtering matters because the TNI engine in Chapter 8 matches skill gaps against the **Skills covered** field of published courses — a catalog that is categorised and tagged well is the single biggest contributor to useful automation later.

3.2 Creating a course (SOP)

Only administrators and HR managers can publish courses. The form captures the content hub in one screen: identity and cost, delivery and assessment, and the module list. Cost is entered in rupees and feeds directly into the Level-4 ROI calculation, so an accurate figure here pays off in analytics later.

1. Click **New Course** (top right).
2. Fill **Course title *** and optionally a **Code** (e.g. POSH-101) and **Provider**.
3. Choose the **category**, **mode** and **level**; set **Duration (hours)**.
4. For external content paste the **External link / content URL**; e-learning modules are defined in the next field instead.
5. Enter **Skills covered** as comma-separated values — these are used for TNI matching, so reuse the skill names your performance reviews use.
6. Write the **Description / learning objectives** employees will read on the card.

7. Tick **Compliance course** (and pick POSH / CYBERSECURITY / SAFETY / DATA_PRIVACY / OTHER) if this is a statutory certification; tick **Promotion-gate course** if it gates a career ladder.
8. Set the **Passing score %** (default 60) and the **Cost** in rupees.
9. In the **Modules** box, add one module per line as *type|title|minutes* where type is video, document, quiz, link, live_session or on_job — e.g. *video|Welcome & policy overview|15*.
10. Click **Publish course**. The course appears in the catalog immediately and becomes assignable, requestable and TNI-matchable.

> **Deletion is cascading**
>
> Deleting a course (trash icon) permanently removes it **and all of its enrolments**. The confirmation dialog states this explicitly. Prefer archiving by editing a course's status when you only want to retire it from assignment.

4. Learning Paths & Promotion Tracks

```
Navigation: Sidebar → Learning & Development → Learning Paths | Route: /learning/paths
```

A learning path bundles courses into an ordered journey. Typical paths are new-joiner onboarding, an annual compliance bundle, or a Team-Lead-to-Manager transition. Each path card shows its category, status, enrolment count and completion percentage, and its sequence as numbered chips (*1. Workplace Safety Induction → 2. Cybersecurity Awareness → … → Path complete*). Optional courses are marked, and required ones are assumed otherwise. When a path is assigned, the enrolment engine creates one enrolment per course in the path — required and optional alike — and stamps each with the path reference so progress can be tracked as a whole.

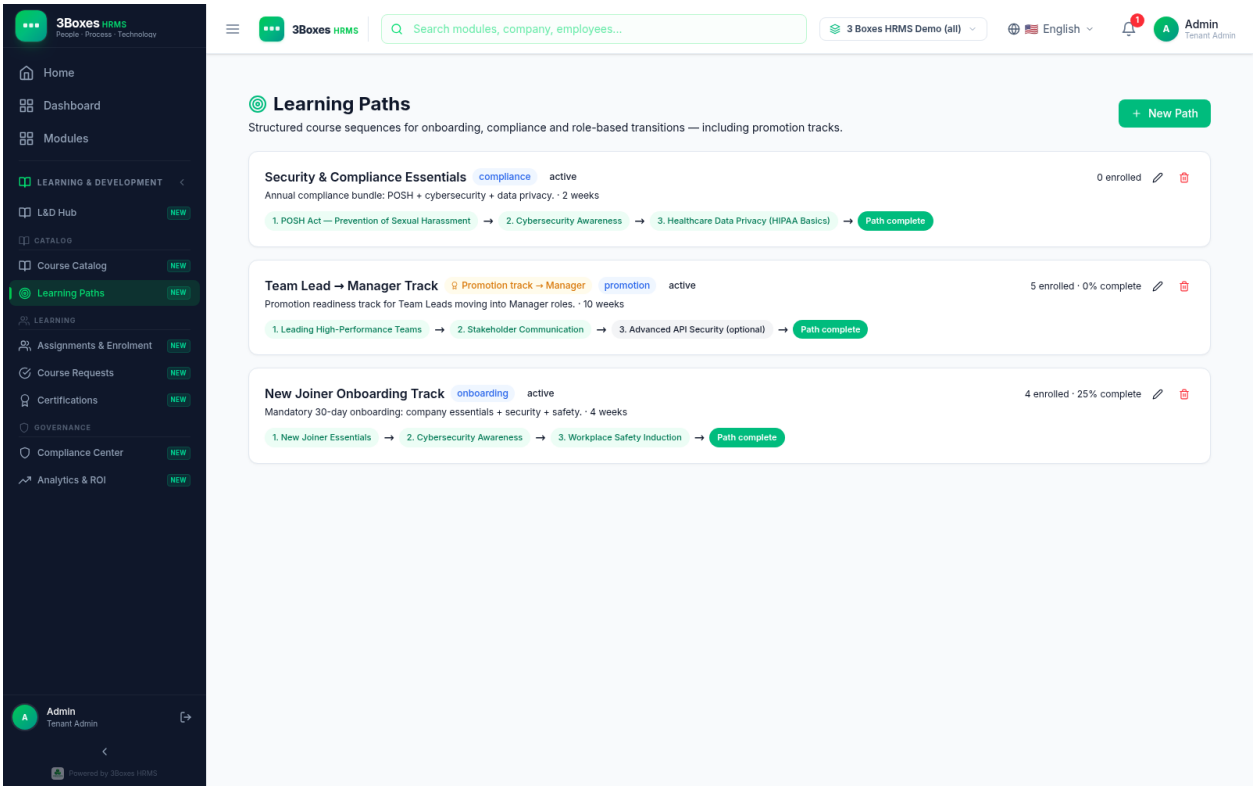


Exhibit 4-1: Learning Paths with a promotion track (Team Lead → Manager)

4.1 Promotion readiness tracks

Ticking **Promotion readiness track** and setting a **Target role** turns the path into a career gate. Promotion-track paths surface in the Analytics page's **Promotion readiness** panel, which counts employees as *eligible* only when they have completed every required course in the track, and *in progress* otherwise. This gives HR an objective, course-completion-based promotion signal instead of an improvised one, and gives employees a visible checklist for the role they are aiming at. The demo tenant ships a *Team Lead → Manager Track* with three courses so you can see the pattern end to end.

4.2 Creating a path (SOP)

1. Click **New Path**.
2. Enter the **Path name *** and choose a category: onboarding, compliance, role transition, promotion or skill upskilling.
3. For promotion tracks set the **Target role** and tick **Promotion readiness track**.
4. Optionally set **Duration (weeks)** and a description; choose status draft or active.
5. Use **+ Add course to path…** to pick courses in the intended order; reorder with the arrow buttons in **Course sequence**.
6. Mark any course as optional if it should not block path completion.
7. Click **Create path** (or **Save changes** when editing). The path becomes assignable from the Assignments page and requestable as a bundle.

> **Path assignment behaviour**
>
> Assigning a path creates one enrolment per course. If an employee already has an enrolment for one of those courses, that enrolment is kept — the engine never creates duplicates. Progress is reported per course, not per path.

5. Assignment & Enrolment Engine

```
Navigation: Sidebar → Learning & Development → Assignments & Enrolment | Route: /learning/enrollments
```

The Assignments page is the operational heart of the module: it is where mandatory and optional training is pushed to people, and where completion is tracked live. Five stat cards summarise the current scope — **Total enrolments**, **assigned**, **in progress**, **completed** and **Overdue** — and filter chips (all, assigned, in progress, completed, failed) narrow the tracking table. Each row shows the employee, course, the **Source** that created the enrolment (assigned, request, manager recommendation or compliance auto), progress bar, due date, status badge and result. The list is role-scoped: administrators see the company, managers see themselves and their direct reports, employees see only their own rows.

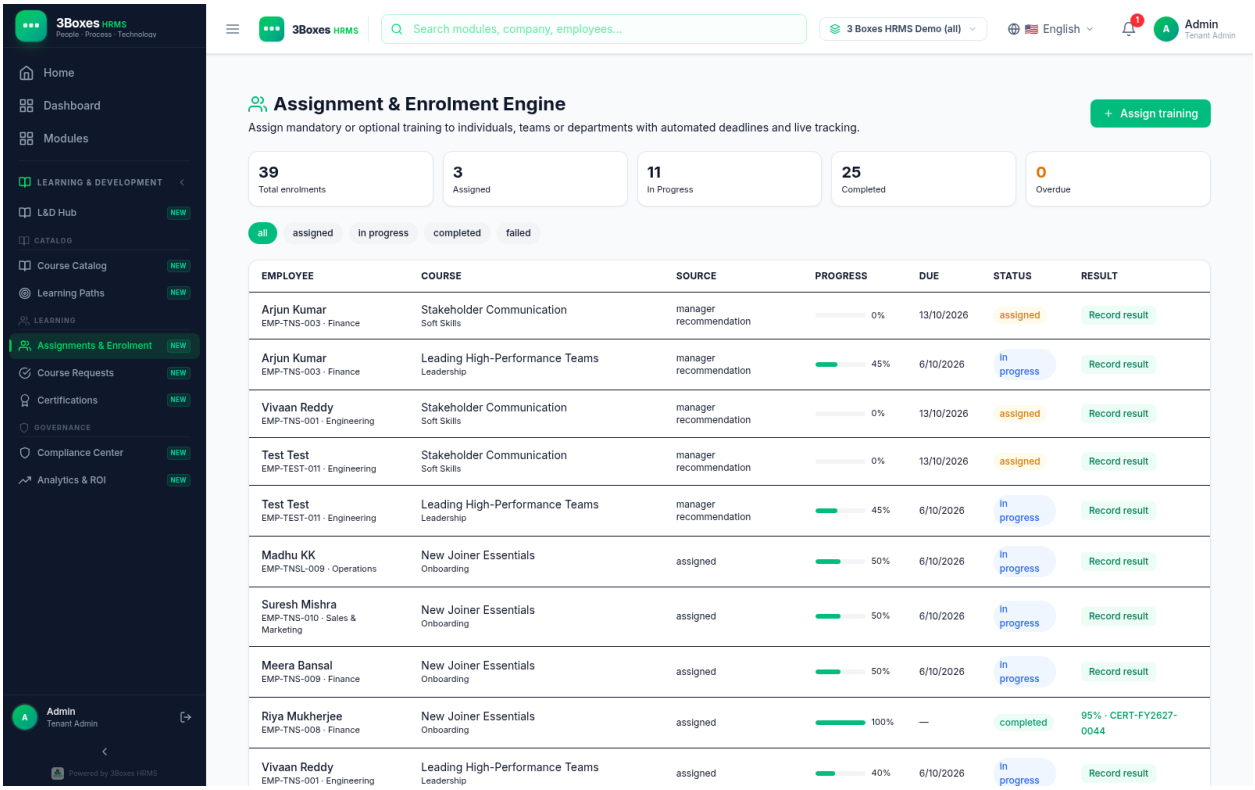


Exhibit 5-1: Assignment & Enrolment Engine — stats, filters and the tracking table

5.1 Assigning training (SOP)

Approvers (admins, HR managers and managers) click **Assign training** to open the assignment wizard. The wizard supports three targeting modes, which cover the vast majority of real training pushes: individuals, a whole department, or the entire company context. Every created enrolment notifies its employee, and the confirmation toast reports exactly what happened, including any duplicates that were skipped.

1. Click **Assign training**.
2. Select the course (or a learning path) from **Select course ***.
3. Choose the assignee type: **individual** (multi-select employees via search), **department** (**Select department ***) or **company** (all active employees in the selected company context).
4. Set the **Due Date** and tick **Mandatory** for statutory or gated training.
5. Optionally note a **Reason** — it is stored on the assignment for audit.
6. Click **Create enrolments**. The toast confirms: *"Assigned — N enrolment(s) created, N notified"*.

> **Duplicate protection**
>
> Enrolments are deduplicated by the platform: one employee can hold only one enrolment per course. Re-assigning skips existing enrolments rather than duplicating them, so bulk pushes are safe to repeat.

5.2 Recording results and lifecycle

Enrolments move through a simple lifecycle: **assigned** → **in progress** → **completed** or **failed**, with **expired** as a terminal state for overdue items. Employees advance their own progress (Start/Continue on the Hub), but the assessment result is an approver action: click **Record result** on the row and enter the assessment score in the prompt. A score at or above the course's passing score (default 60%) completes the enrolment, records the certificate automatically (Chapter 7) and closes any matching skill gap (Chapter 8). Below the pass mark the enrolment is marked failed and the attempt count increments, so the employee can be re-assigned for a retake.

The result cell shows the outcome at a glance: a percentage plus the certificate number for passes (e.g. *95% · CERT-FY2627-0044*), or the **Record result** button while an assessment is outstanding. Recording also captures pre-test and post-test scores when provided, which feed the Level-2 learning analytics — the module defaults the pre-test to 40 when a trainer does not run a separate baseline, and the prompt shows the pass threshold so there is no guessing.

6. Course Nomination & Approval

`Navigation: Sidebar → Learning & Development → Course Requests | Route: /learning/requests`

Not all training should be pushed top-down. The Course Requests page gives employees a structured way to ask for development, and gives supervisors and L&D administrators a budget-conscious approval queue. The page's subtitle captures the contract: employees request courses or HR assigns mandatory paths, and reviewers weigh budget and team priorities. Every request stores who asked, what for, how much it is expected to cost in rupees, and why — so approval decisions are auditable months later.

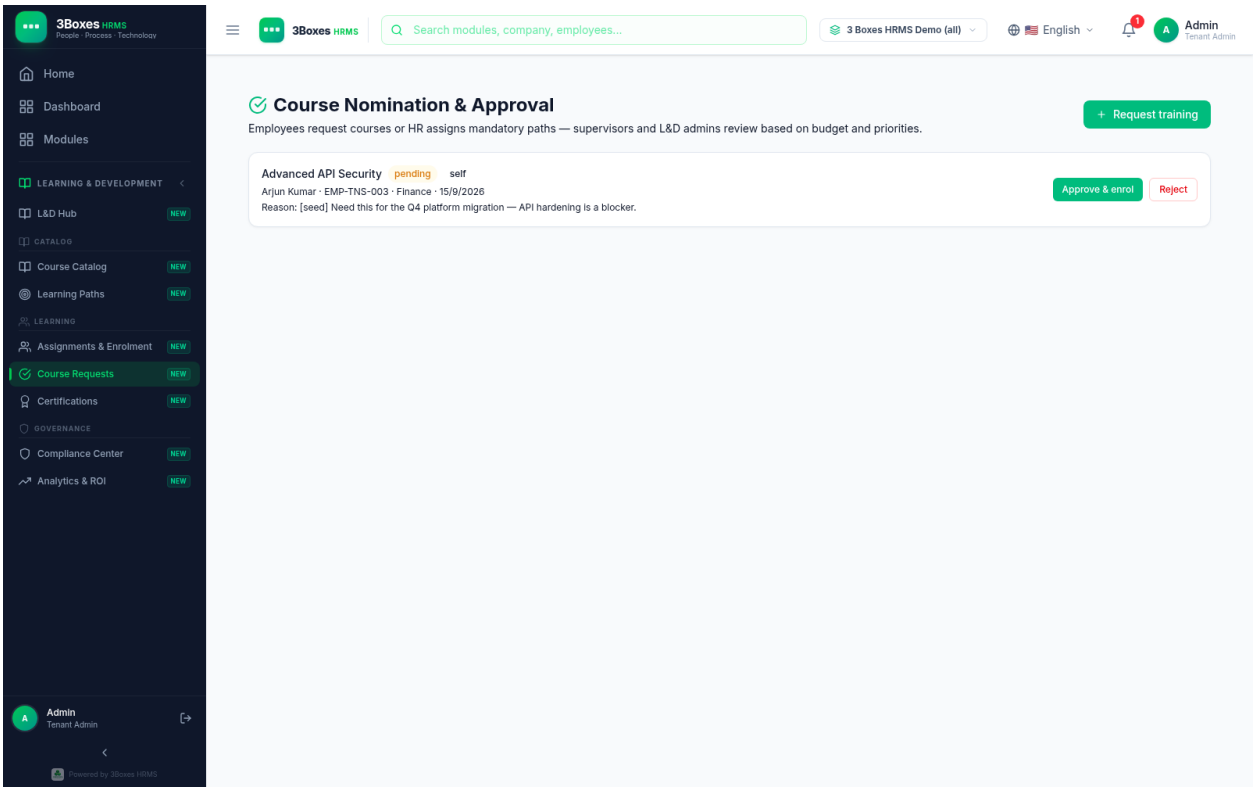


Exhibit 6-1: Course Nomination & Approval — request form and review queue

6.1 Submitting a request (SOP)

1. Click **Request training** (visible to every signed-in user).
2. Pick **Select a course from the catalog**, or type **Or an external course title** for training that is not in the catalog yet.
3. Choose the request type: **Self nomination** or **Manager suggested**.
4. Enter the **Expected cost (₹)** — this is the number your reviewer sees against budget.
5. Explain the driver in the reason box: *"Why do you need this training? (skill gap, project need, career goal…)"*.
6. Click **Submit request**. The toast confirms your supervisor / L&D admin will review it, and the reporting manager is notified automatically.

Requests appear instantly to the right reviewers: HR managers and administrators see company-wide requests, managers see their own and their direct reports'. Pending rows show the request type badge, the expected cost, and the reason text, so a reviewer can usually decide without a follow-up meeting.

6.2 Approving or rejecting (SOP)

1. Open **Course Requests** and locate the pending row.
2. Click **Approve & enrol** and optionally type an approval note.
3. The platform auto-enrols the employee in the requested course (duplicates are skipped safely) and notifies them — the toast reads *"Approved — employee auto-enrolled"*.
4. For unsuitable requests click **Reject** and enter a **Rejection reason**; the reason is stored and visible on the request row.

> **One-click approval**
>
> Approval always creates the enrolment in the same action. There is no second step where an approver must remember to enrol someone — this removes the classic "approved but never enrolled" failure mode of spreadsheet processes.

7. Tracking, Completion & Certifications

```
Navigation: Sidebar → Learning & Development → Certifications | Route:
/learning/certifications
```

Certifications are the module's proof of competence. When an approver records a passing result, the platform issues a digital certificate automatically — no template work, no manual emailing. Each certificate carries a unique financial-year number in the format **CERT-FY2627-NNNN** (the FY starts in April), the course title, the employee, the score, the issue date and a validity window. Non-compliance certificates are issued with lifetime validity; compliance certifications are valid for **12 months** so statutory re-certification can never be forgotten.

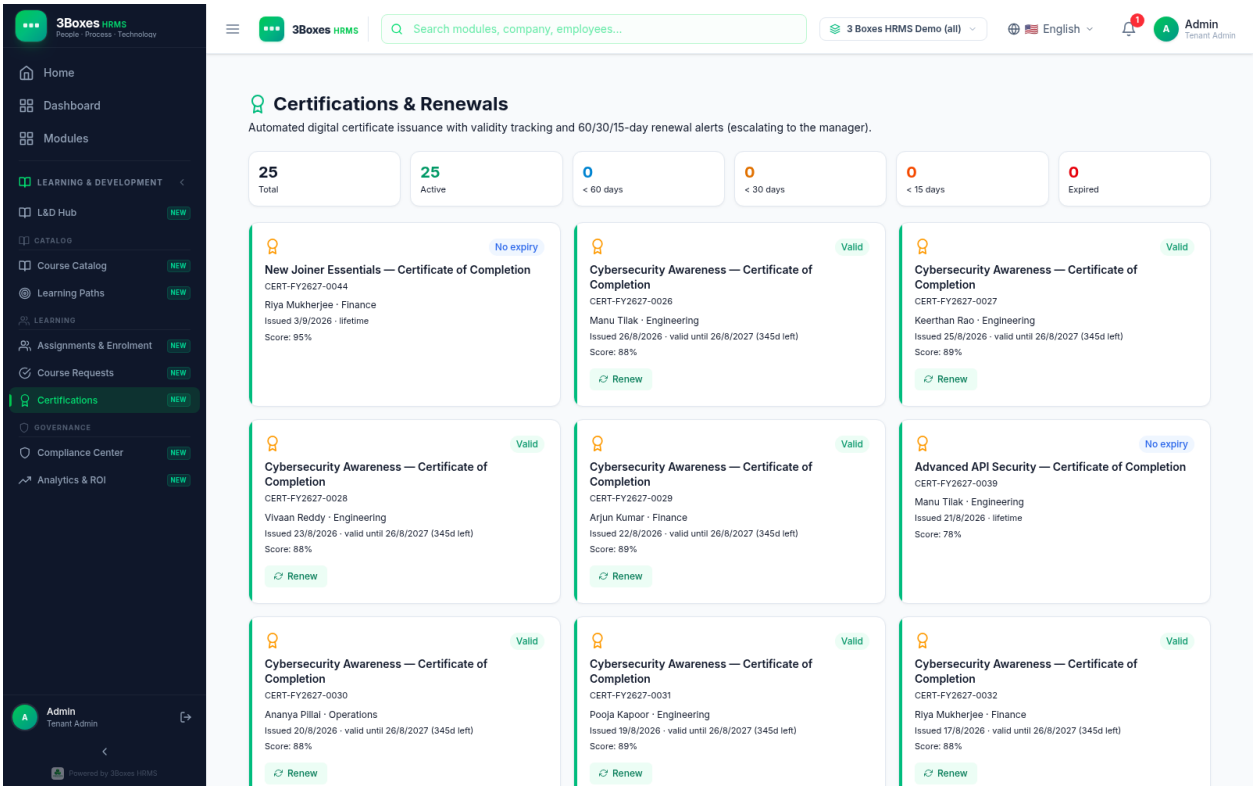


Exhibit 7-1: Certifications & Renewals — KPI bands and certificate cards

7.1 The six validity bands

The page opens with six KPI cards — **Total**, **Active**, **< 60 days**, **< 30 days**, **< 15 days** and **Expired** — and every certificate card wears a band badge matching the same vocabulary: **Valid**, **No expiry**, **Expiring < 60d**, **Expiring < 30d**, **Expiring < 15d** or **Expired**. Cards also show the days remaining (e.g. *345d left*) so urgency is always explicit. The daily compliance engine (Chapter 9 and Chapter 11) sends renewal alerts as each band is entered, escalating to the direct manager at the 15-day mark.

Band	Trigger	What happens
Valid	More than 60 days to expiry	No action; card shows days remaining.
Expiring < 60d	60 days or fewer remaining	First renewal alert to the employee (once).
Expiring < 30d	30 days or fewer remaining	Second renewal alert to the employee (once).
Expiring < 15d	15 days or fewer remaining	Final alert, **escalated to the reporting manager**.
Expired	Validity end date passed	Status flips to expired automatically; renew to reactivate.
No expiry	Lifetime certificate	Never alerts; Renew is hidden.

Table 7-1: Certificate validity bands and alert behaviour

7.2 Renewing a certificate (SOP)

1. Open **Certifications** and filter attention to the amber and red band cards.
2. Have the employee re-complete the assessment (assign the course again if needed).
3. As an administrator, click **Renew** on the certificate card.
4. The platform extends validity by 12 months (or the course's renewal period) from the later of the current expiry or today, resets the alert counters, and returns the certificate to **Valid**.

Renewal is deliberately a one-click administrative action rather than a new enrolment flow: the historical certificate, its number and its audit trail are preserved, and only the validity window moves. Employees can filter to their own certificates with the **mine** view, which is what the My Learning header count on the Hub reads from.

8. TNI — Skill Gaps from Performance Reviews

Training Needs Identification (TNI) is what makes this module a development system rather than a course player. The module reads completed performance reviews and, when a manager has rated a skill below the threshold or flagged it for improvement, creates a **Skill Gap** record for that employee. Each gap stores the skill name, the current level (e.g. *5/10*), the required level (threshold + 1), the source review, and — when the catalog contains a course whose **Skills covered** tags match — the auto-suggested course. Gaps then travel a visible lifecycle: **open → suggested → in training → closed**.

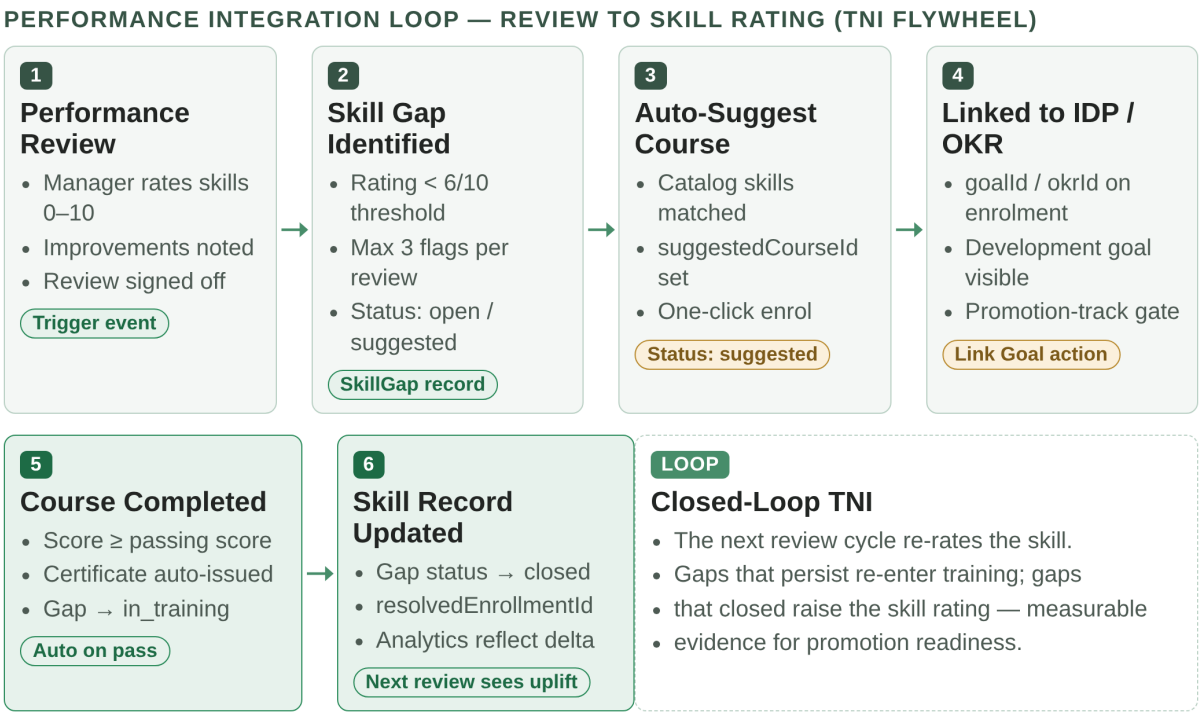


Exhibit 8-1: The performance-integration loop — review to skill rating, automatically

The loop in Exhibit 8-1 is the module's core promise to business leaders: training spend is aimed at measured gaps, and completion feeds back into the next review cycle. Managers see open gaps surfaced on the L&D Hub (*"N open skill gap(s) — From performance reviews (TNI)"*), and L&D administrators can generate gaps on demand after an appraisal cycle via the skill-gap API (*action: generate*).

8.1 How gaps are generated

- **Rating threshold** — any review skill rated below 6 out of 10 (configurable) becomes a gap, with current level recorded as "rating/10".
- **Improvement flags** — the review's free-text improvements are scanned for phrases like "needs improvement", "improve", "gap" or "weak"; up to three flags per review become gaps.
- **Duplicate protection** — an existing open, suggested or in-training gap for the same employee and skill is never duplicated.
- **Auto-match** — the engine searches published courses whose skills tags overlap the gap; a match sets the gap to *suggested* with the course attached, otherwise it stays *open* for manual course selection.

8.2 Acting on a gap (SOP)

1. Review the gap list (L&D Hub alert card, or the skill-gap API list view).
2. Accept the auto-suggested course, or attach a different course to the gap.
3. Enrol directly from the gap — the gap moves to **in training** and links the enrolment so completion can be traced back to the original review.
4. When the employee passes, the gap closes automatically and the enrolment reference is stored on the gap record as evidence.

Development goals are first-class too: any enrolment can be linked to an employee's individual development plan (IDP) or an OKR through the **Link Goal** action, which stores the goal and OKR references on the enrolment. This is how a promotion-track path, a skill gap and a quarterly objective can all point at the same course and report progress through one record — the exact integration promised in the module's design brief.

9. Compliance Tracking & Policy Sign-Offs

```
Navigation: Sidebar → Learning & Development → Compliance Center | Route:
/learning/compliance
```

The Compliance Center is the audit-facing half of the module. It answers three questions continuously: who is due for mandatory training, who has signed what, and can you prove both in one click. The header row shows **Overall compliance (all departments)** as a percentage plus the top departments with progress bars (e.g. Engineering 48%, Finance 67%), computed live from the requirement matrix. An **Export audit report** button downloads a CSV with one row per employee and Complete/Overdue/N-A per requirement — the artefact most auditors ask for first.

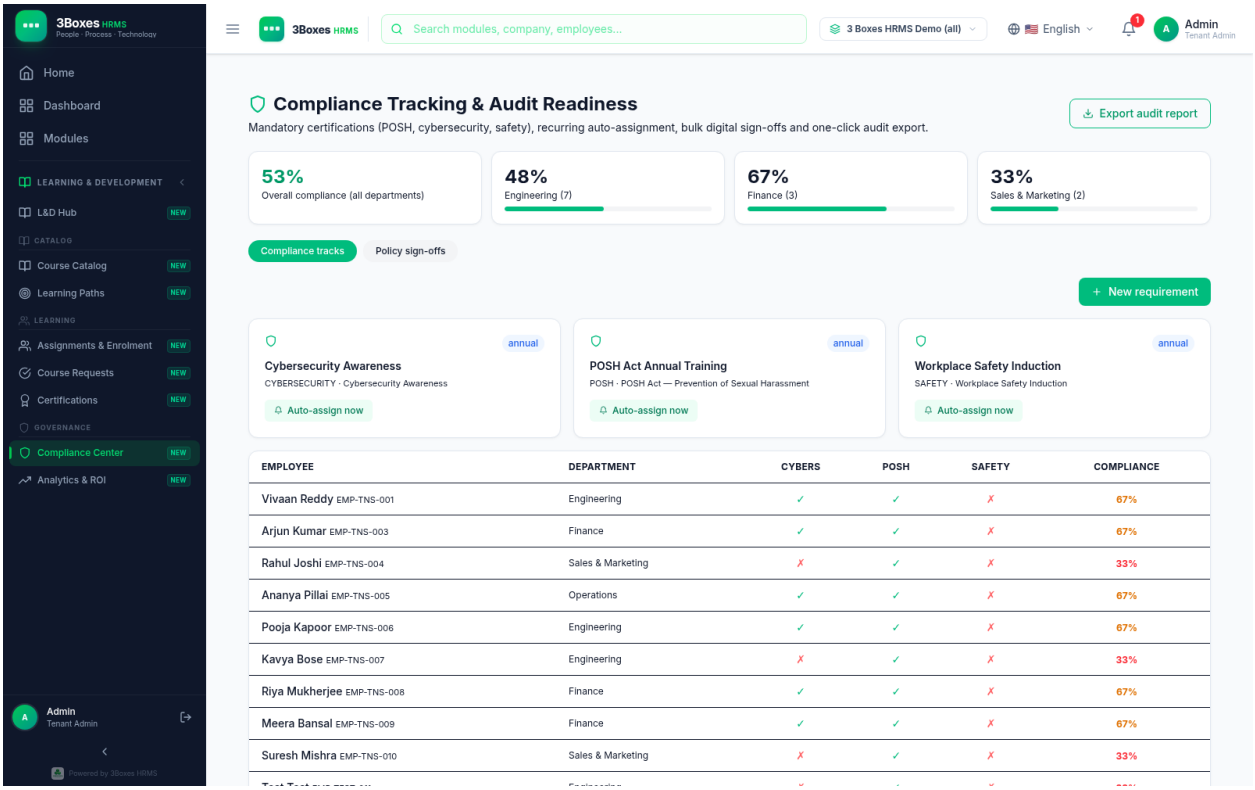


Exhibit 9-1: Compliance Center — compliance KPIs, requirement cards and the employee matrix

9.1 Compliance tracks

A compliance requirement binds a category (POSH, cybersecurity, safety, data privacy, healthcare or other) to a catalog course and a frequency: one-time, annual, biannual or quarterly. The matrix below the cards shows every employee, their department, a check-mark column per requirement and a personal compliance percentage. The daily engine evaluates the cycle window per frequency — quarterly means the last three months, annual the last twelve — and any employee without a completed enrolment inside the window is marked overdue.

1. Click **New requirement**.
2. Name it (e.g. *POSH Act Training 2026*), pick the **category** and **frequency**.
3. Link the compliance course with **Link compliance course…**.
4. Click **Create**, then use **Auto-assign now** on the card whenever you want to push it: the engine enrols all active employees as mandatory with a due date 30 days out and notifies them.

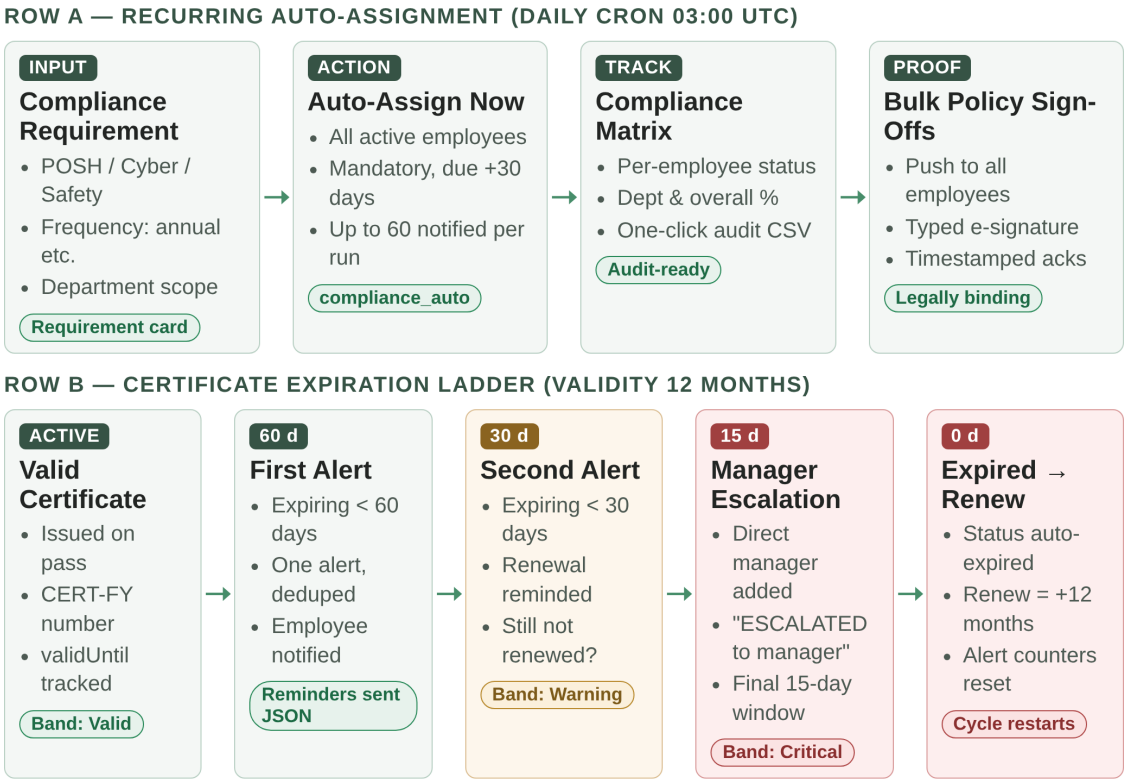


Exhibit 9-2: The compliance engine — recurring auto-assignment above, the 60/30/15-day expiration ladder below

9.2 Bulk policy sign-offs

The second tab, **Policy sign-offs**, handles legally binding acknowledgements that are not courses: codes of conduct, handbooks, policy updates. Pushing a campaign creates a pending acknowledgement for every active employee; each employee types their full name to sign, and the platform stores the signature name and timestamp. Campaign cards track *"24/40 signed (60%) · due date"* with a progress bar, a per-user **Sign now** action, a **Remind pending** broadcast, and **Close** for finalised campaigns. Because signatures are timestamped per employee per policy version, the campaign list itself is acceptable audit evidence.

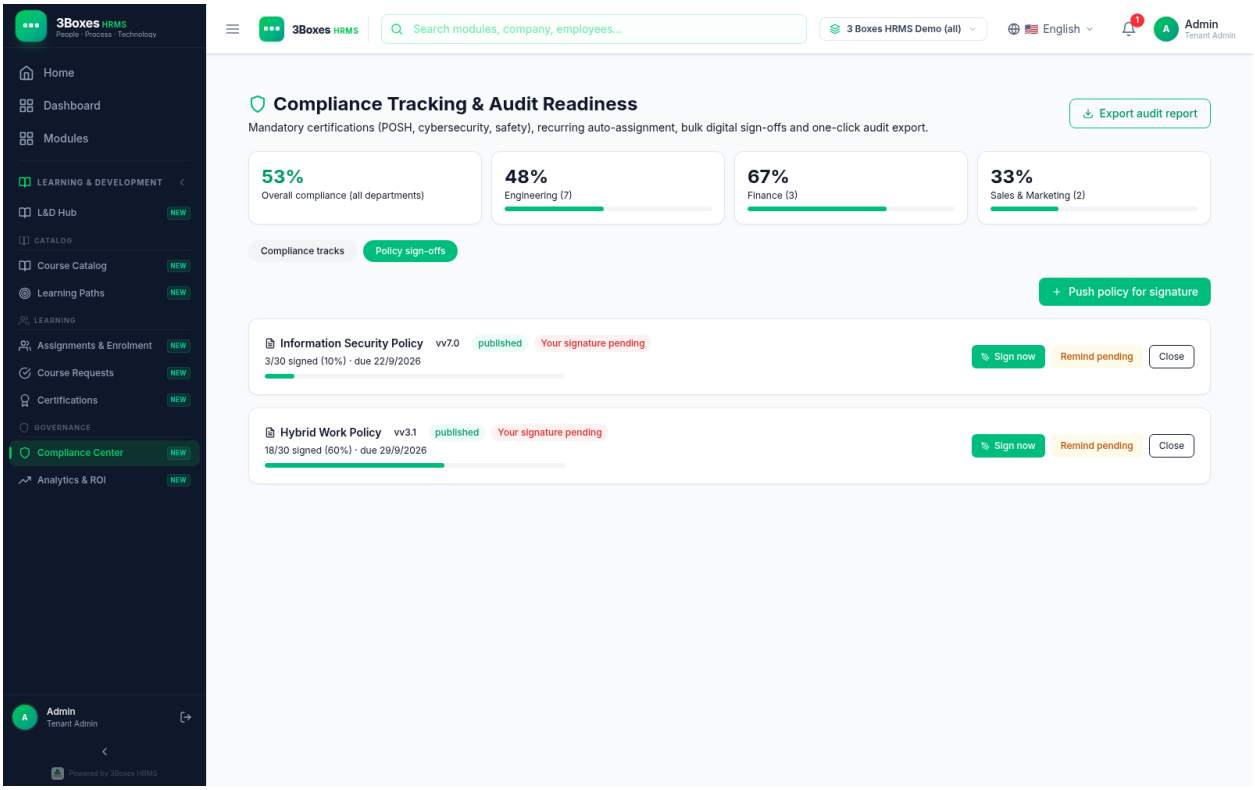


Exhibit 9-3: Policy sign-offs tab — campaign progress, Sign now, Remind and Close actions

10. Analytics & ROI — Kirkpatrick Model

```
Navigation: Sidebar → Learning & Development → Analytics & ROI | Route:
/learning/analytics
```

Training that cannot be measured will always be the first line item cut. The Analytics page therefore implements the Kirkpatrick four-level evaluation model end to end, topped with four headline KPIs: **Completion rate**, **Total enrolments**, **Training ROI** (with spend and headcount) and **Knowledge gain (L2)**. In the demo tenant the sample data yields a 64% completion rate, 39 enrolments, a 607% ROI on ₹1,88,400 of spend, and a +39.1 point knowledge gain — numbers that make an L&D budget conversation concrete.

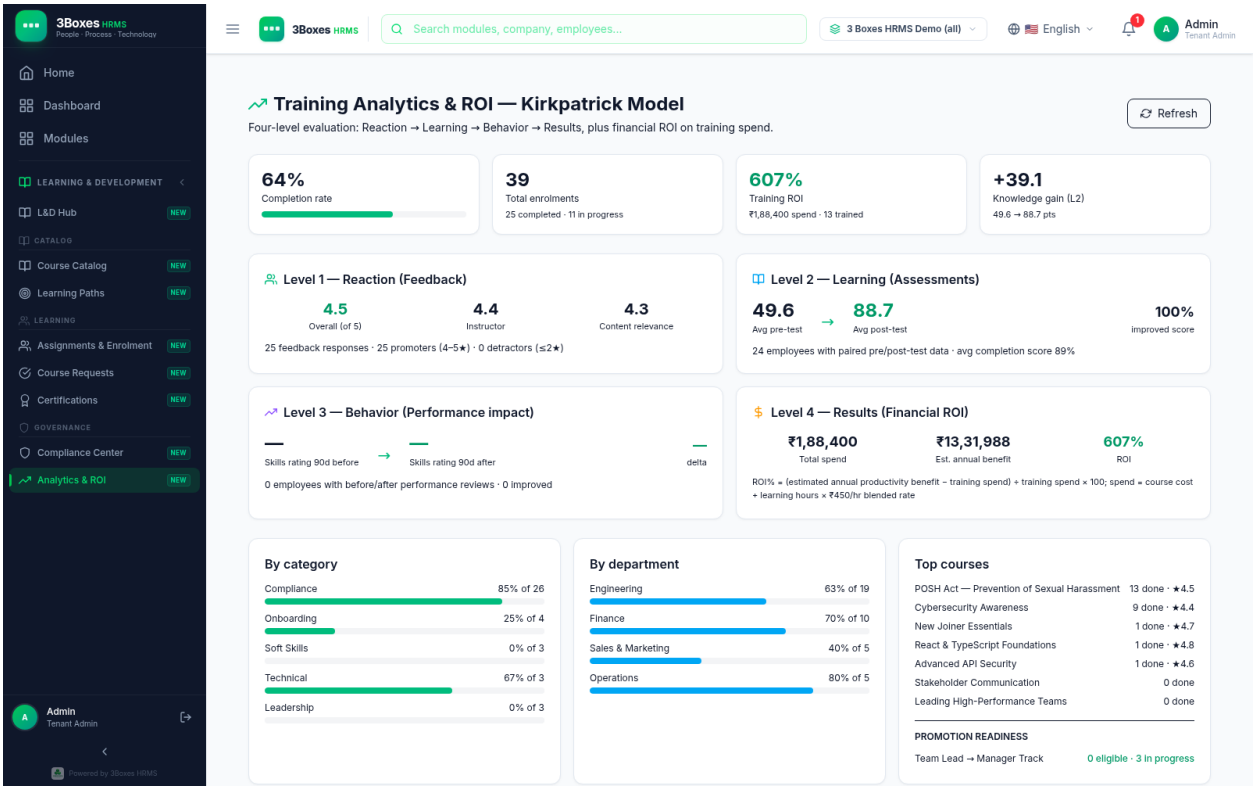


Exhibit 10-1: Training Analytics & ROI — Kirkpatrick levels, rollups and promotion readiness

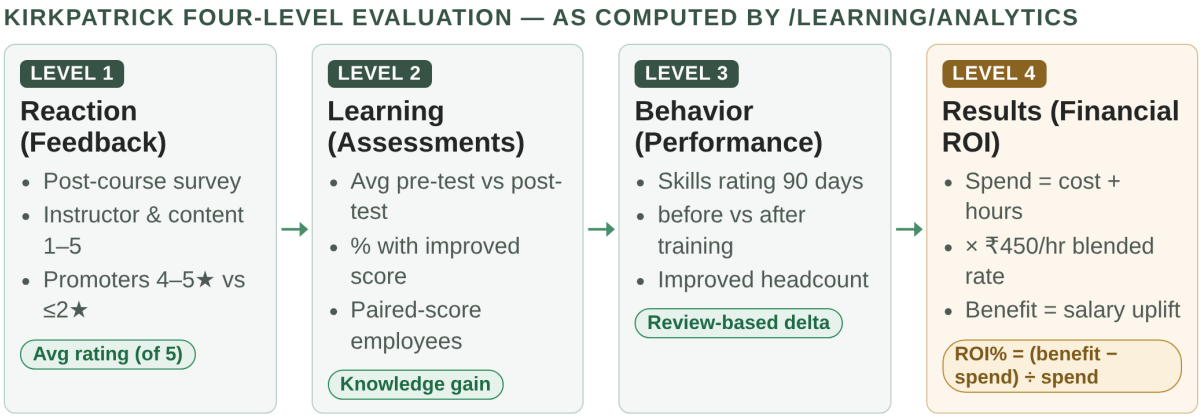


Exhibit 10-2: What each Kirkpatrick level measures and how it is computed

10.1 The four levels in practice

- **Level 1 — Reaction.** Post-course feedback captured on completion: overall rating of five, instructor rating and content-relevance rating, plus promoter (4–5 star) and detractor (≤2 star) counts.
- **Level 2 — Learning.** Paired pre-test and post-test averages with the percentage of employees who improved, computed only for enrolments that have both scores.
- **Level 3 — Behavior.** The employee's performance-review skills rating in the 90 days before versus the 90 days after course completion, reported as a delta with the number of employees who improved. This is

behaviour change measured in the organisation's own review instrument.

- **Level 4 — Results.** Financial ROI: total spend is course cost plus learning hours valued at a ₹450/hour blended rate; estimated annual benefit applies a productivity uplift of 10% of salary per ten points of assessed improvement (capped at 15%) against a default ₹6,00,000 annual salary; ROI = (benefit − spend) ÷ spend × 100. The exact formula string is printed under the card so finance can challenge the assumptions.

10.2 Rollups and promotion readiness

Beneath the level cards, three rollup panels break the same data down by category, by department (percentage of that department's employees trained) and the top courses by completions and average rating. The **Promotion readiness** panel lists every promotion-track path with the count of employees eligible (all required courses complete) versus in progress — turning Chapter 4's career gates into a live talent pipeline view. Together these panels are what an HR leadership review needs; export them via browser print or screenshot for board packs.

11. Notifications & Automation

The L&D module plugs into the platform's multi-channel notification center (module key: *learning*). Four events cover the lifecycle, and every event supports the standard four channels — in-app, email, SMS and Slack/Teams webhooks — with editable subject and body templates that use the same double-brace placeholders as the rest of the platform, e.g. **{{Course_Name}}**, **{{Due_Date}}**, **{{Recipient_Name}}**. Tenant administrators can override any template in the notification console (*Notifications → Templates & Rules*) without code changes.

Event	Fires when	Primary recipients
learning.training.assigned	A new assignment or approved request creates enrolments; also used for compliance reminders.	Enrolled employee (up to 60 per push)
learning.compliance.reminder	Daily cron: certificate renewal bands, overdue mandatory enrolments, and sign-off reminders.	Employee; manager on 15-day escalation
learning.training.completed	An enrolment is completed with a passing score.	Employee and reporting manager
learning.manager.completion_report	Manager-level completion summaries.	Reporting manager

Table 11-1: Learning notification events and their triggers

Beyond notifications, three automations run without any user involvement. First, the duplicate-proof enrolment engine (one enrolment per employee per course) makes bulk assignment idempotent. Second, the auto-issue chain on passing: certificate created, matching skill gap closed, employee and manager notified. Third, the daily compliance cron at 03:00 UTC which scans active certificates for the 60/30/15-day bands, marks expired certificates, and nudges overdue mandatory enrolments.

11.1 The compliance cron in detail

- **Deduped alerts.** Each certificate stores which alert thresholds have already fired (a JSON map of 60/30/15 timestamps), so employees receive exactly one alert per band — renewal resets the map.
- **Escalation.** When a certificate enters the 15-day band, the alert is marked "ESCALATED to manager" and the direct reporting manager is added as recipient.
- **Auto-expiry.** Certificates past their validity date flip to expired automatically; the renewal action of Chapter 7 brings them back.
- **Overdue nudges.** Mandatory enrolments past their due date remind the employee each run until completed or expired.

12. Roles & Permissions Matrix

Access follows the platform's five standard roles. Two internal role groups drive the UI: **administrators** (super admin, tenant admin, admin, HR manager) unlock the management consoles, and **approvers** (administrators plus managers) unlock assignment, approval and result-recording actions. The module itself is visible to all five roles, subject to the tenant's module whitelist. Use the matrix below when setting up role assignments or investigating a "button missing" report.

Capability	Employee	Manager	HR Mgr	Admin
View My Learning & own certificates	Yes	Yes	Yes	Yes
Request training / nominate courses	Yes	Yes	Yes	Yes
Sign policy acknowledgements	Yes	Yes	Yes	Yes
Approve / reject course requests	No	Yes (team)	Yes (all)	Yes (all)
Assign training (individual / dept / company)	No	Yes	Yes	Yes
Record assessment results	No	Yes	Yes	Yes
Create / edit courses and paths	No	No	Yes	Yes
Delete courses and paths	No	No	No	Yes
Renew certificates	No	No	Yes	Yes
Create compliance requirements / auto-assign	No	No	Yes	Yes
Push policy sign-off campaigns / remind / close	No	No	Yes	Yes
Export audit report (CSV)	No	No	Yes (view)	Yes
View company-wide analytics & ROI	No	Team view	Yes	Yes

Table 12-1: L&D permission matrix by role

> **Scope notes**
>
> "Yes (team)" means a manager sees and acts on their own requests and their direct reports' only; company-wide queues are reserved for HR managers and admins. The server enforces this scoping on every API call, not just in the UI.

13. SOP Quick Reference

This chapter condenses the procedures of Chapters 2–9 into three role playbooks. Each line is an action in the UI, in order, and can be used as a desk aid or onboarding cheat-sheet for the module. Times assume the demo tenant's seeded catalog; a live tenant's steps are identical.

13.1 L&D administrator playbook

1. **Build the catalog:** Course Catalog → New Course → fields + modules → Publish course. Tag skills accurately; set realistic passing scores and rupee costs.
2. **Structure journeys:** Learning Paths → New Path → add courses in order → mark promotion tracks with a target role.
3. **Define mandates:** Compliance Center → New requirement → category + frequency + linked course → Auto-assign now each cycle.
4. **Push policies:** Compliance Center → Policy sign-offs → Push policy for signature → monitor signed percentage → Remind pending → Close.
5. **Clear queues:** Hub alert cards → Course Requests approvals; generate TNI gaps after each appraisal cycle; enrol or assign from suggested gaps.
6. **Keep certifications healthy:** Certifications → work the <60/<30/<15-day bands → Renew after re-assessment.
7. **Prove it:** Compliance Center → Export audit report before every audit or management review; Analytics & ROI for the Kirkpatrick story.

13.2 Manager playbook

1. **Identify:** after each performance review, note skills rated below the bar — the TNI engine turns them into suggested courses automatically.
2. **Nominate:** Course Requests → Request training → Manager suggested → cost + reason.
3. **Approve:** Course Requests → review team requests → Approve & enrol (or Reject with a reason).
4. **Assign:** Assignments & Enrolment → Assign training → team individuals → due date + Mandatory.
5. **Close the loop:** record results from the tracking table; watch your completion report notifications; use Link Goal to tie enrolments to IDPs/OKRs.
6. **Escalation partner:** respond to 15-day certificate escalations — either the employee re-certifies now, or you schedule the retake.

13.3 Employee playbook

1. **Check in:** L&D Hub → My Learning → Start/Continue the nearest-due course.
2. **Grow:** browse the Course Catalog; Request training with a cost estimate and reason when you find something valuable.
3. **Complete:** finish modules, take the assessment, view your result via the tracking table or your certificate.
4. **Stay compliant:** complete POSH / cybersecurity / safety before their due dates; renewal alerts arrive at 60, 30 and 15 days.
5. **Sign:** Compliance Center → Policy sign-offs → Sign now by typing your full name — your acknowledgement is timestamped.

14. FAQ & Troubleshooting

Q1. Why can I not see the Assign training button?

Assignment, approval and result recording are approver actions — administrators, HR managers and managers. If you are an employee, request training instead via Course Requests; if you are a manager and still cannot see it, ask your tenant admin to check your role assignment and the tenant's module whitelist for *learning*.

Q2. An employee was assigned a course twice — is that a bug?

No. Enrolments are deduplicated by the platform: one employee holds at most one enrolment per course. Re-assigning skips the existing enrolment and the toast's "created vs notified" counts reflect that. The employee simply keeps their current progress.

Q3. Why did a certificate get 12-month validity while another is lifetime?

Validity follows the course type at completion time: compliance courses (POSH, cybersecurity, safety, data privacy or other statutory content) issue 12-month certificates so statutory re-certification recurs; all other courses issue lifetime certificates. The Renew action extends a certificate by 12 months from the later of its expiry or today.

Q4. The renewal alert fired twice for the same certificate.

The daily cron deduplicates per threshold band via the certificate's reminder map, so one alert per band is the expected behaviour. If you re-created a certificate or renewed it, counters reset by design. Persistent duplicates usually indicate two certificates for the same course (e.g. one from an earlier cycle) — revoke the stale one from the certifications API or renew only the current one.

Q5. My assessment was recorded as failed but the employee answered correctly.

The pass mark is the course's **Passing score %** (default 60). Check the course edit form; if the threshold was set unusually high (e.g. 90), lower it and record the result again — attempts are logged, so the history remains

visible.

Q6. Can employees record their own scores?

No. Result recording is deliberately approver-only to keep assessments honest. Employees control progress (Start/Continue), and approvers control the verdict. The server rejects self-recording even if the endpoint is called directly.

Q7. How do I train the whole company at once?

Use **Assign training** → assignee type **company**: every active employee in the selected company context is enrolled (duplicates skipped). For recurring statutory training prefer a compliance requirement with **Auto-assign now**, which also sets a mandatory 30-day due date and keeps the audit matrix updated.

Q8. Where does the ROI number come from?

Level 4 uses the formula printed under the card: spend = course cost + learning hours × ₹450/hour blended rate; benefit = a productivity uplift of 10% of salary per ten points of assessed improvement, capped at 15%, on a default ₹6,00,000 annual salary; ROI% = (benefit − spend) ÷ spend × 100. Treat it as a directional business case and recalibrate the blended rate with finance if your cost base differs.

Q9. Policy sign-off says "Your signature pending" — what must I do?

Open Compliance Center → Policy sign-offs and click **Sign now** on the campaign card, then type your full name exactly as your legal signature. The platform stores the typed name and timestamp as a legally binding acknowledgement for that policy version.

Q10. Why is a skill gap not suggesting any course?

Auto-matching compares the gap's skill with each published course's **Skills covered** tags. If no course shares the tag, the gap stays *open*. Either add the skill tag to a suitable course (Catalog → Edit) or attach a course to the gap manually before enrolling.

Q11. Do learning paths track a single progress percentage?

Progress is tracked per course enrolment; the path card shows the aggregate percentage across its enrolments. Employees complete the sequence course by course, and optional courses never block path completion in the promotion-readiness count.

Q12. What happens to enrolments when a course is deleted?

Deleting a course cascades: all of its enrolments and their certificates are removed. That is why the confirm dialog names the course explicitly. For retirements, prefer editing the course's status to archived so history is preserved.

15. Glossary

Term	Meaning in 3Boxes HRMS L&D
Blended rate	The per-hour cost used to value learning time in ROI (₹450/hour by default), combining course cost with employee hours spent.
Compliance requirement	A recurring mandate (one-time/annual/biannual/quarterly) binding a category such as POSH to a catalog course.
Compliance track	The requirement card + matrix view that shows who is complete, overdue or not applicable per mandate.
Content hub	The course-level collection of videos, documents, quizzes, live sessions, external links and on-the-job tasks.
Course module	One unit of course content, defined as type|title|minutes in the catalog form (video, document, quiz, link, live_session, on_job).
Enrolment	The single employee-course record carrying progress, due date, scores, feedback and result; unique per employee per course.
Enrolment source	Why the enrolment exists: assigned, self, request, manager_recommendation or compliance_auto.
Escalation (15-day)	The final renewal alert band; the direct reporting manager is added as recipient alongside the employee.
IDP / OKR link	Optional references stored on an enrolment tying it to an employee's development plan or a company objective.
Kirkpatrick model	Four-level training evaluation: Reaction, Learning, Behavior, Results — implemented on the Analytics page.
Level 3 (Behavior)	Skills-rating comparison in the 90 days before vs after completion, from performance reviews.
Passing score	The percentage at or above which an assessment passes (default 60%).
Policy sign-off	A bulk digital-signature campaign; acknowledgements are typed signatures with timestamps per employee per version.
Promotion-gate course	A course flagged as mandatory for a career ladder; promotion tracks surface eligibility from its completion.
Promotion readiness	Analytics panel counting employees eligible (all required path courses complete) vs in progress per track.
Renewal band	The 60/30/15-day windows before certificate expiry that trigger deduplicated alerts.
Skill gap (TNI)	A record of a skill shortfall with current/required level, source review and suggested course; lifecycle open → suggested → in training → closed.
TNI	Training Needs Identification — deriving training demand from performance reviews, supervisor feedback and compliance mandates.

Term	Meaning in 3Boxes HRMS L&D
Training request	An employee's nomination for a catalog or external course, with expected cost and reason, pending approval.
Validity window	A certificate's validFrom–validUntil range; 12 months for compliance courses, lifetime otherwise.

Table 15-1: L&D glossary

For platform-wide terms — tenants, roles, approval tiers, notification channels — refer to the core 3Boxes HRMS documentation set. The notification console's template editor documents every placeholder available to learning events, and the compliance export format is stable for downstream GRC tooling. Together with this manual, those references complete the operating picture for a production-ready learning organisation.